

MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

September 21, 2025

Outline

- 1 Real Analysis Lecture 8
 - Open Balls and Open Sets

Outline

- 1 Real Analysis Lecture 8
 - Open Balls and Open Sets

Euclidean space

Euclidean space

n -dimensional **euclidean space** is

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Definitions:

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Definitions:

Given $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Definitions:

Given $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$

Ⓐ $\vec{x} + \vec{y} = (x_1 + y_1, \dots, x_n + y_n)$

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Definitions:

Given $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$

- (a) $\vec{x} + \vec{y} = (x_1 + y_1, \dots, x_n + y_n)$
- (b) $a\vec{x} = (ax_1, \dots, ax_n)$

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Definitions:

Given $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$

- (a) $\vec{x} + \vec{y} = (x_1 + y_1, \dots, x_n + y_n)$
- (b) $a\vec{x} = (ax_1, \dots, ax_n)$
- (c) $\vec{x} - \vec{y} = \vec{x} + (-1)\vec{y} = (x_1 - y_1, \dots, x_n - y_n)$

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Definitions:

Given $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$

- (a) $\vec{x} + \vec{y} = (x_1 + y_1, \dots, x_n + y_n)$
- (b) $a\vec{x} = (ax_1, \dots, ax_n)$
- (c) $\vec{x} - \vec{y} = \vec{x} + (-1)\vec{y} = (x_1 - y_1, \dots, x_n - y_n)$
- (d) $\vec{0} = (0, \dots, 0)$

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Definitions:

Given $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$

- (a) $\vec{x} + \vec{y} = (x_1 + y_1, \dots, x_n + y_n)$
- (b) $a\vec{x} = (ax_1, \dots, ax_n)$
- (c) $\vec{x} - \vec{y} = \vec{x} + (-1)\vec{y} = (x_1 - y_1, \dots, x_n - y_n)$
- (d) $\vec{0} = (0, \dots, 0)$
- (e) $\vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i$

Euclidean space

n -dimensional **euclidean space** is

$$\mathbb{R}^n = \{(a_1, a_2, \dots, a_n) : a_1, \dots, a_n \in \mathbb{R}\}.$$

Definitions:

Given $\vec{x} = (x_1, \dots, x_n)$ and $\vec{y} = (y_1, \dots, y_n)$

- (a) $\vec{x} + \vec{y} = (x_1 + y_1, \dots, x_n + y_n)$
- (b) $a\vec{x} = (ax_1, \dots, ax_n)$
- (c) $\vec{x} - \vec{y} = \vec{x} + (-1)\vec{y} = (x_1 - y_1, \dots, x_n - y_n)$
- (d) $\vec{0} = (0, \dots, 0)$
- (e) $\vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i$
- (f) $|\vec{x}| = (\vec{x} \cdot \vec{x})^{1/2} = (\sum_{i=1}^n x_i^2)^{1/2}$

Metric properties

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Theorem

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Theorem

- (a) *(positivity) $|\vec{x}| \geq 0$ with equality iff $\vec{x} = \vec{0}$*

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Theorem

- (a) (positivity) $|\vec{x}| \geq 0$ with equality iff $\vec{x} = \vec{0}$
- (b) (symmetry) $|\vec{x} + \vec{y}| = |\vec{y} + \vec{x}|$

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Theorem

- (a) *(positivity)* $|\vec{x}| \geq 0$ with equality iff $\vec{x} = \vec{0}$
- (b) *(symmetry)* $|\vec{x} + \vec{y}| = |\vec{y} + \vec{x}|$
- (c) *(triangle inequality)* $|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|$

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Theorem

- (a) *(positivity)* $|\vec{x}| \geq 0$ with equality iff $\vec{x} = \vec{0}$
- (b) *(symmetry)* $|\vec{x} + \vec{y}| = |\vec{y} + \vec{x}|$
- (c) *(triangle inequality)* $|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|$

It also satisfies

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Theorem

- (a) *(positivity)* $|\vec{x}| \geq 0$ with equality iff $\vec{x} = \vec{0}$
- (b) *(symmetry)* $|\vec{x} + \vec{y}| = |\vec{y} + \vec{x}|$
- (c) *(triangle inequality)* $|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|$

It also satisfies

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Theorem

- (a) *(positivity)* $|\vec{x}| \geq 0$ with equality iff $\vec{x} = \vec{0}$
- (b) *(symmetry)* $|\vec{x} + \vec{y}| = |\vec{y} + \vec{x}|$
- (c) *(triangle inequality)* $|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|$

It also satisfies

- (a) *(scaling)* $|c\vec{x}| = |c| |\vec{x}|$

Metric properties

The norm $|\vec{x}|$ is an example of a **metric**.
It satisfies several important properties:

Theorem

- (a) (positivity) $|\vec{x}| \geq 0$ with equality iff $\vec{x} = \vec{0}$
- (b) (symmetry) $|\vec{x} + \vec{y}| = |\vec{y} + \vec{x}|$
- (c) (triangle inequality) $|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|$

It also satisfies

- (a) (scaling) $|c\vec{x}| = |c| |\vec{x}|$
- (b) (Cauchy-Schwartz) $|\vec{x} \cdot \vec{y}| \leq |\vec{x}| |\vec{y}|$

Open sets

An **open ball** of radius r centered at $\vec{a}$ is

Open sets

An **open ball** of radius r centered at $\vec{a}$ is

$$B(\vec{a}; r) = \{\vec{x} : |\vec{x} - \vec{a}| < r\}.$$

Open sets

An **open ball** of radius r centered at $\vec{a}$ is

$$B(\vec{a}; r) = \{\vec{x} : |\vec{x} - \vec{a}| < r\}.$$

Definition

Open sets

An **open ball** of radius r centered at $\vec{a}$ is

$$B(\vec{a}; r) = \{\vec{x} : |\vec{x} - \vec{a}| < r\}.$$

Definition

A point $\vec{a}$ in a subset $A \subseteq \mathbb{R}^n$ is called an **interior point** of A if there exists $r > 0$ with $B(\vec{a}; r) \subseteq A$.

Open sets

An **open ball** of radius r centered at $\vec{a}$ is

$$B(\vec{a}; r) = \{\vec{x} : |\vec{x} - \vec{a}| < r\}.$$

Definition

A point $\vec{a}$ in a subset $A \subseteq \mathbb{R}^n$ is called an **interior point** of A if there exists $r > 0$ with $B(\vec{a}; r) \subseteq A$. If every point of A is an interior point, then A is called an **open set**.

Challenge!

Problem

Prove that the empty set $\emptyset$ and the whole space $\mathbb{R}^n$ are open.

Challenge!

Problem

Let $\vec{a} \in \mathbb{R}^n$. Show that the singleton set

$$A = \{\vec{a}\}$$

is not open.

Challenge!

Problem

Let $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$. Prove that the **open rectangle**

$$(a, b) \times (c, d) = \{(x, y) : a < x < b, \ c < y < d\}$$

is an open set

Challenge!

Problem

Prove that an open ball is an open set.